

SUPPLEMENTAL MATERIAL

Effects of Electronic Cigarettes on Indoor Air Quality and Health

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Supplemental Table 1. Summary of PM_{2.5} and UFP levels, background concentration, emission protocol, room size, and air exchange rate for studies in Figure 2

a. PM_{2.5}

Study	Location	Mean, µg/m ³	Background, µg/m ³	Emission Protocol	Room Size, m ³	Air Exchange Rate, h ⁻¹
E-Cig Aerosols						
Melstrom et al. (2017)	Laboratory room	1121 ± 2430	18 ± 15	3 volunteers use e-cigs for 2 h	53	5.0
Nguyen et al. (2019)	6 vape shops	766 ± 1442	12 ± 6	An average of 3.9 puffs per min generated by e-cig users	168-323	0.1-4.8
Soule et al. (2017)	An e-cig vaping event in a hotel	607 ± 212	3 ± 1	A 2-day e-cig event with 59-86 active e-cig users	4023	NA
Volesky et al. (2018)	Laboratory room	290	2	1 e-cig user generated 7 puffs with 4-s puffing duration	38	NA
Zhao et al. (2017)	Laboratory room	188 ± 433	8 ± 1	1 volunteer used e-cigs with 3-s puffing duration, 3-s hold, and 24s pause for 10 min	80	4.1
Schober et al. (2013)	Laboratory room	197 ± 193	6	3 volunteers used e-cigs for 2 h	45	0.6
Czogala et al. (2014)	Laboratory room	152 ± 87	32 ± 30	5 dual volunteers used e-cigs twice for 5 min with a 30-min interval	39	2.6
Ruprecht et al. (2017)	Laboratory room	ND	0	Volunteers used e-cigs for 2 to 3 h; one puff per min for about 7 min, followed by 3 min hold	48	1.5
T-Cig Aerosols						
Forster et al. (2018)	Laboratory room	1129 ± 466	6 ± 4	Volunteers smoked a puff every 30 s for 8 puffs.	38	1.2-7.7

Czogala et al. (2014)	Laboratory room	819 ± 229	32 ± 30	5 volunteers smoked twice for 5 min with a 30-min interval	39	2.6
Kungsjuniti et al. (2015)	Smoking rooms in international airports	533	14	NA	NA	NA
Ruprecht et al. (2017)	Laboratory room	500 ± 128	0	Volunteers smoked for about 7 min with 3-min intervals	48	1.5
Loffredo et al. (2016)	15 cafes and 5 restaurants allowed smoking	433 ± 217	72-81	T-cigs and waterpipes smoking observed at all sampling locations	128-235	NA
Höllbacher et al. (2017)	Laboratory room	374	0	1 volunteer smoked t-cigs for 30 min	30	0.5
Acevedo-Bolton et al. (2014)	Laboratory room	267	3-8	1 volunteer smoked 1 or 2 t-cigs for 5-10 min	121 or 460	NA
Movsisyan et al. (2014)	A hospital before smoking ban	222	NA	High frequency of smoking t-cigs observed in multiple locations in the hospital before smoking ban	NA	NA
Zhou et al. (2016)	A smoking-permitted area (i.e., employee break room) in a casino	164	23	25-43 active smokers in the smoking-permitted area in the casino	NA	NA
Shamo et al. (2015)	78 restaurants before smoking ban	126 ± 109	12 ± 23	20% smoking prevalence, 2.6 active smokers per restaurant, 1.11 burning t-cigs per 100 m ³	NA	NA
Kim et al. (2016a)	62 hospitality venues (i.e., restaurants and bars) before smoking ban	109 ± 2	46 ± 2	Smoking observed in 25 hospitality venues	90-1680	NA

Scheitel et al. (2016)	Smoking in a car	105 ± 21	2	1 volunteer smoked 1 or 2 t-cigs in a car	NA	NA
Kim et al. (2016b)	75 bars before smoking ban	98 ± 2	25 ± 2	Smoking was observed in 68-96% of the 75 sampled bars before smoking ban	NA	NA
Buettner-Schmidt et al. (2015)	135 hospitality venues (i.e., bars and restaurants)	88 ± 122	9	0.3 burning t-cigs per 100 m ³ , smoking observed in 42% of the hospitality venues	494 ± 601	NA
Liu et al. (2014)	114 restaurants and bars	70 ± 124	NA	3.7 % smoking prevalence, 0.3 active smokers observed per 100 m ³	NA	4.5 (range: 0-14)

b. UFPs

Study	Location	Mean, particles/cm ³	Background, particles/cm ³	Emission Protocol	Room Size, m ³	Air Exchange Rate, h ⁻¹
E-Cig Aerosols						
Schober et al. (2014)	Laboratory room	61682 ± 17524	4466	3 volunteers used e-cigs for 2 h	45	0.6
Nguyen et al. (2019)	6 vape shops	60667 ± 52296	14580 ± 7845	an average of 3.9 puffs per min generated by e-cig users	168-323	0.1-4.8
Melstrom et al. (2017)	Laboratory room	34961 ± 27992	3707 ± 823	3 volunteers used e-cigs for 2 h	53	5.0
Zhao et al. (2017)	Laboratory room	24800 ± 21400	6390 ± 158	1 volunteer used e-cigs with 3-s puffing duration, 3-s hold, and 24s pause for 10 min	80	4.1
Ruprecht et al. (2017)	Laboratory room	8657 ± 560	0	Volunteers used e-cigs for 2 to 3 h; one puff per min for about 7 min, followed by 3 min hold	48	1.5

Scungio et al. (2018)	Laboratory room	7690 ± 169	NA	1 volunteer used e-cigs for 10 min at 0.7-1.5 puffs/min	40	0.2
Volesky et al. (2018)	Laboratory room	7169	1389	1 e-cig user generated 7 puffs with 4-s puffing duration	38	NA
T-Cig Aerosols						
Forster et al. (2018)	Laboratory room	210000 ± 75500	4297 ± 4639	Volunteers smoked a puff every 30 s for 8 puffs.	38	1.2-7.7
Ruprecht et al. (2017)	Laboratory room	122672 ± 37327	0	Volunteers smoked for about 7 min with 3-min intervals	48	1.5
Scheitel et al. (2016)	Smoking in a car	113998 ± 34382	15545	1 volunteer smoked 1 or 2 t-cigs in a car	NA	NA
Konstantopoulou et al. (2014)	A bar/restaurant before smoking ban	27172	12509	An average of 4 smokers observed	NA	NA

Supplemental Table 2. List of reviewed chemical pollutants for both mainstream and secondhand aerosols and corresponding references

Chemicals	References	
	Mainstream Aerosols	Secondhand Aerosols
Vegetable Glycerin	Dai et al. (2017); Geiss et al. (2015); Margham et al. (2016); Zhao et al. (2018)	Geiss et al. (2015); Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Propylene Glycol	Dai et al. (2017); Geiss et al. (2015); Margham et al. (2016); Zhao et al. (2018)	Geiss et al. (2015); Johnson et al. (2018); Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Nicotine	Czogala et al. (2014); Dai et al. (2017); El-Hellani et al. (2018); Farsalinos et al. (2018b); Geiss et al. (2015); Goniewicz et al. (2013); Goniewicz et al. (2018); Margham et al. (2016); Pagano et al. (2016); Sleiman et al. (2016); Son et al. (2018); Talih et al. (2016); Tayyarah et al. (2014); Zhao et al. (2018)	Ballbè et al. (2014); Chen et al. (2017); Czogala et al. (2014); Geiss et al. (2015); Johnson et al. (2018); Liu et al. (2017); Melstrom et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Formaldehyde	Farsalinos et al. (2017); Farsalinos et al. (2018a); Farsalinos et al. (2018c); Flora et al. (2016); Flora et al. (2017); Geiss et al. (2015); Gillman et al. (2016); Goniewicz et al. (2014); Jensen et al. (2015); Khlystov et al. (2016); Kosmider et al. (2014); Kosmider et al. (2018); Margham et al. (2016); Ogunwale et al. (2017); Qu et al. (2018); Qu et al. (2019); Salamanca et al. (2018); Samburova et al. (2018); Sleiman et al. (2016); Talih et al. (2016); Talih et al. (2017); Tayyarah et al. (2014)	Johnson et al. (2018); Liu et al. (2017); Ruprecht et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Acetaldehyde	Farsalinos et al. (2018a); Farsalinos et al. (2018c); Flora et al. (2016); Flora et al. (2017); Geiss et al. (2015); Gillman et al. (2016); Goniewicz et al. (2014); Khlystov et al. (2016); Kosmider et al. (2014); Kosmider et al. (2018); Margham et al. (2016); Ogunwale et al. (2017); Qu et al. (2018); Qu et al. (2019); Samburova et al. (2018); Sleiman et al. (2016); Talih et al. (2016); Talih et al. (2017); Tayyarah et al. (2014)	Johnson et al. (2018); Liu et al. (2017); Ruprecht et al. (2017); Schober et al. (2014); Schripp et al. (2013)

Propanal	Farsalinos et al. (2018a); Farsalinos et al. (2018c); Geiss et al. (2015); Goniewicz et al. (2014); Khlystov et al. (2016); Kosmider et al. (2014); Kosmider et al. (2018); Margham et al. (2016); Ogunwale et al. (2017); Qu et al. (2018); Qu et al. (2019); Samburova et al. (2018); Sleiman et al. (2016); Talih et al. (2016); Talih et al. (2017); Tayyarah et al. (2014)	Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Acrolein	Farsalinos et al. (2018a); Farsalinos et al. (2018c); Flora et al. (2016); Flora et al. (2017); Geiss et al. (2015); Gillman et al. (2016); Goniewicz et al. (2014); Khlystov et al. (2016); Kosmider et al. (2014); Kosmider et al. (2018); Margham et al. (2016); Ogunwale et al. (2017); Qu et al. (2018); Qu et al. (2019); Samburova et al. (2018); Sleiman et al. (2016); Talih et al. (2016); Talih et al. (2017); Tayyarah et al. (2014)	Johnson et al. (2018); Liu et al. (2017); Ruprecht et al. (2017); Schober et al. (2014)
Acetone	Farsalinos et al. (2018a); Geiss et al. (2015); Goniewicz et al. (2014); Khlystov et al. (2016); Kosmider et al. (2014); Kosmider et al. (2018); Margham et al. (2016); Ogunwale et al. (2017); Qu et al. (2018); Qu et al. (2019); Samburova et al. (2018); Sleiman et al. (2016); Talih et al. (2016); Talih et al. (2017);	Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Benzene	Flora et al. (2016); Margham et al. (2016); Pankow et al. (2017); Sleiman et al. (2016); Tayyarah et al. (2014)	Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Toluene	Flora et al. (2016); Goniewicz et al. (2014); Margham et al. (2016); Tayyarah et al. (2014)	Liu et al. (2017); Schober et al. (2014); Schripp et al. (2013)
Chromium	Margham et al. (2016); Tayyarah et al. (2014); Williams et al. (2013)	Liu et al. (2017); Saffari et al. (2014); Schober et al. (2014)
Aluminum	Palazzolo et al. (2017); Williams et al. (2013)	Saffari et al. (2014); Schober et al. (2014)
Copper	Lerner et al. (2015); Margham et al. (2016); Palazzolo et al. (2017); Williams et al. (2013)	Saffari et al. (2014); Schober et al. (2014)
Zinc	Margham et al. (2016); Palazzolo et al. (2017); Williams et al. (2013)	Saffari et al. (2014); Schober et al. (2014)
Cadmium	Flora et al. (2016); Goniewicz et al. (2014); Margham et al. (2016); Palazzolo et al. (2017); Tayyarah et al. (2014)	Liu et al. (2017); Saffari et al. (2014); Schober et al. (2014)

Nickel	Goniewicz et al. (2014); Margham et al. (2016); Mikheev et al. (2016); Palazzolo et al. (2017); Tayyarah et al. (2014); Williams et al. (2013)	Liu et al. (2017); Saffari et al. (2014); Schober et al. (2014)
Lead	Goniewicz et al. (2014); Margham et al. (2016); Palazzolo et al. (2017); Tayyarah et al. (2014); Williams et al. (2013)	Saffari et al. (2014); Schober et al. (2014)
Iron	Margham et al. (2016); Palazzolo et al. (2017); Williams et al. (2013)	Saffari et al. (2014); Schober et al. (2014)
Arsenic	Flora et al. (2016); Margham et al. (2016); Mikheev et al. (2016); Palazzolo et al. (2017); Tayyarah et al. (2014)	Liu et al. (2017); Schober et al. (2014)
N'-Nitrosonornicotine (NNN)	Farsalinos et al. (2015); Flora et al. (2016); Goniewicz et al. (2014); Margham et al. (2016); Tayyarah et al. (2014)	NA
4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK)	Farsalinos et al. (2015); Flora et al. (2016); Goniewicz et al. (2014); Margham et al. (2016); Tayyarah et al. (2014)	NA